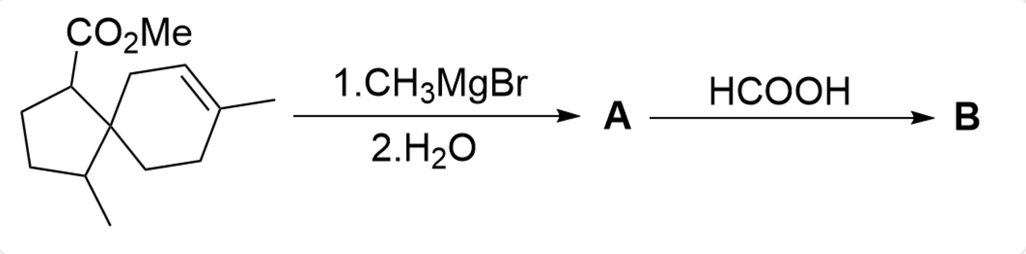
Question

Spiro compounds are often used in the synthesis of complex polycyclic compounds. The following figure describes a reaction of a spiro compound:



The image describes an organic multi-step reaction. The substrate is CC1CCC(C(OC)=O)C12CCC(C)=CC2, which reacts with C[Mg]Br and then water is added to generate **A**. **A** reacts with O=CO to generate **B**

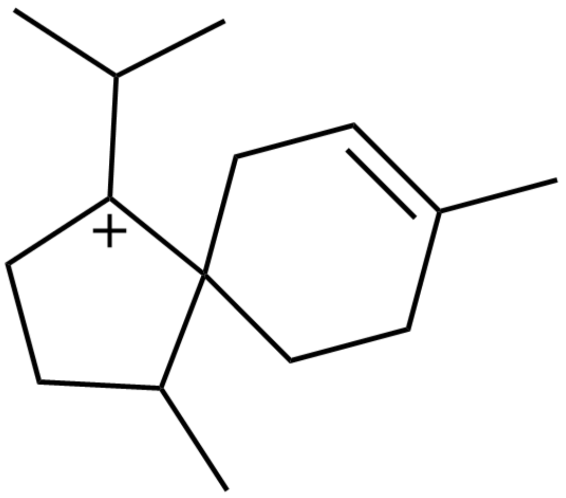
Known: **B** contains two five-membered rings and one six-membered ring.

Which of the following statements is correct:

A. **B** contains 5 chiral carbon atoms.

B. **B** contains two unsaturation bonds.

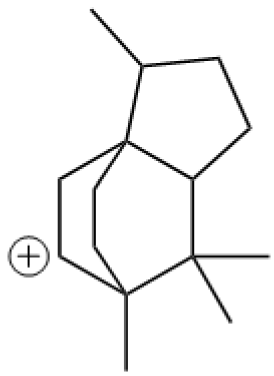
C.



CC(C)[C+]1C2(CCC(C)=CC2)C(C)CC1

The above image is the key intermediate for the generation of **B**

D.



CC1CCC(C12CCC([CH+]C2)3C)C3(C)C

The above image shows the key intermediate for generating **B**.

E. **B** is completely oxidized to **C** by acidic potassium permanganate solution, then the chemical formula of **C** is $C_{15}H_{24}O_3$

F. All other options are incorrect.

Answer

Correct Answer: E

Detailed Explanation

The reaction involving the addition of methyl Grignard reagent is relatively straightforward, as the Grignard reagent reacts with an ester to yield a tertiary alcohol.

CHECKPOINT 1 PTS

Grignard reagent reacts with ester to form tertiary alcohol

Thus, the structure of **A** is

CC1CCC(C(C)(O)C)C12CCC(C)=CC2.

CHECKPOINT 1 PTS

Structure of **A** is CC1CCC(C(C)(O)C)C12CCC(C)=CC2

Tertiary alcohols readily undergo dehydration in a strongly acidic environment such as formic acid, forming a stable tertiary carbocation. Therefore, the intermediate leading to **B** includes CC1CCC([C+](C)C)C12CCC(C)=CC2.

CHECKPOINT 1 PTS

Tertiary alcohol dehydrates in formic acid to form tertiary carbocation

CHECKPOINT 1 PTS

Intermediate of **B** includes CC1CCC([C+](C)C)C12CCC(C)=CC2

B contains three rings, indicating an intramolecular cyclization reaction. The carbocation, being a strong electrophile, can only react intramolecularly with the remaining alkene as the nucleophile, leading to ring formation.

CHECKPOINT 1 PTS

Alkene nucleophilically attacks carbocation, forming a ring intramolecularly.

Given that **B** contains two five-membered rings and one six-membered ring, while **A** already has a spiro ring system (five + six), the newly formed ring must be a five-membered ring. Consequently, the carbon linking the alkene and the carbocation must be a secondary carbon, resulting in the tricyclic intermediate CC1CCC(C(C)2C)C13CC[C+](C)C2C3.

CHECKPOINT 1 PTS

Newly formed ring is a five-membered ring, so the alkene-linked carbon must be secondary

CHECKPOINT 1 PTS

Tricyclic intermediate formed: CC1CCC(C(C)2C)C13CC[C+](C)C2C3

This intermediate eliminates a hydride ion, following the rule of forming a stable alkene, resulting in an endocyclic double bond. Thus, **B** is CC1CCC(C12CC=C(C3C2)C)C3(C)C.

CHECKPOINT 2 PTS

B is CC1CCC(C12CC=C(C3C2)C)C3(C)C

B contains four chiral carbons and one unsaturated bond, so options A and B are incorrect.

CHECKPOINT 1 PTS

B has four chiral carbons and one unsaturated bond, options A and B are incorrect.

The endocyclic double bond in **B** is oxidized by potassium permanganate, cleaving the double bond to form a ketone carbonyl and an aldehyde carbonyl, with the latter further oxidized to a carboxylic acid. The oxidation product has the structure CC1CCC2C(C)(C(CC21CC(O)=O)C(C)=O)C.

CHECKPOINT 1 PTS

Oxidation product structure: CC1CCC2C(C)(C(CC21CC(O)=O)C(C)=O)C

Its molecular formula is $C_{15}H_{24}O_3$, so option E is correct.

CHECKPOINT 0.5 PTS

Oxidation product molecular formula: $C_{15}H_{24}O_3$, option E is correct.

The intermediate in option C is derived from CC1CCC([C+](C)C)C12CCC(C)=CC2 after a single migration, forming a carbocation. This intermediate is more prone to rearrangement, converting the spiro ring to a fused ring. If the alkene were to nucleophilically attack, the resulting product would have excessive ring strain and would not match the description of **B**, so the tricyclic product cannot form, making option C incorrect.

CHECKPOINT 1 PTS

Intermediate in C tends to rearrange, converting spiro to fused ring, unable to form tricyclic product, so C is incorrect

Option D represents the product of a tertiary carbocation CC1CCC([C+](C)C)C12CCC(C)=CC2 reacting with the alkene to form a secondary carbocation. However, if the nucleophilic attack occurs at the other carbon, a more stable tertiary carbocation CC1CCC(C(C)2C)C13CC[C+](C)C2C3 forms instead. Thus, it is not the key intermediate, making option D incorrect.

CHECKPOINT 1 PTS

Carbocation nucleophilic attack favors forming a more stable carbocation: CC1CCC(C(C)2C)C13CC[C+](C)C2C3

CHECKPOINT 0.5 PTS

Option D is not the key intermediate, so D is incorrect

Therefore, option E is correct.

CHECKPOINT 0.5 PTS

Therefore, option E is correct.